

Analysis Report — Kadane's Algorithm (Partner's Code)

This report analyzes the Kadane's algorithm implementation that your partner (Student B) would provide. It covers algorithm description, proofs of correctness, time/space complexity ($\Theta/O/\Omega$), code-review feedback, and empirical validation suggestions.

Algorithm Overview and Correctness

Kadane's algorithm computes the maximum subarray sum in a single pass. It maintains two variables: `maxEndingHere` and `maxSoFar`. The algorithm updates `maxEndingHere` as $\max(x, \text{maxEndingHere} + x)$ and `maxSoFar` as $\max(\text{maxSoFar}, \text{maxEndingHere})$. To track positions, maintain a start candidate index and update on replacements.

Complexity Analysis

Time complexity: $\Theta(n)$ in best, average, and worst cases (single pass). Space complexity: $O(1)$ auxiliary memory. For mathematical justification, each element is processed once performing constant-time operations; hence total operations proportional to n .

Code Review & Optimization Suggestions

Common issues to check in partner's code: - Ensure use of primitive ints to avoid autoboxing. - Validate handling of empty arrays (decide on return value) and all-negative arrays. - Track indices carefully: when starting new subarray set candidate start index. - Avoid unnecessary array accesses inside loop and extra allocations.

Suggested optimizations: - Use primitive long if risk of integer overflow; otherwise keep int. - Add unit tests for edge cases, property-based tests comparing to brute-force for small n. - For benchmarking, run repeats and warm-ups; use median elapsed time.

Empirical Validation Plan

To validate complexity: generate test arrays of sizes $n = 100, 1k, 10k, 100k$. For each size, run 10 warm-up runs then 30 measured runs. Record elapsed times and compute median. Plot time vs n on log-log axes to confirm slope ~ 1 for linear complexity.

Conclusion

Kadane's algorithm is optimal (linear time, constant space) for maximum subarray. The main focus of review should be correctness around indices, handling corner cases, and benchmarking methodology. Include the test harness and CSV output in the submission for reproducibility.